

Objectives

Familiar with Git commands like git init, git status, git add, git commit, git push, and git pull.

In this hands-on lab, you will learn how to

- Setup your machine with Git Configuration
- Integrate notepad++.exe to Git and make it a default editor
- Add a file to source code repository

Prerequisites

- Install Git Bash client in your machine

Notes*:

Please follow the below steps for creating a free account in GitHub.
Don't use cognizant credentials to login to GitHub.

Estimated time to complete this lab: **30 minutes**.

Step 1: Setup your machine with Git Configuration

To create a new repository, signup with GitLab and register your credentials

Login to GitLab and create a "GitDemo" project

1. To check if Git client is installed properly: Open Git bash shell and execute

```
$ git version  
git version 2.21.0.windows.1
```

If output shows Git with its version information that indicates, that Git Client installs properly.

2. To configure user level configuration of user ID and email ID execute

```
$ git config --global user.name "username"  
$ git config --global user.email "username@cognizant.com"
```

3. To check if the configuration is properly set, execute the following command.

```
$ git config --global --list
user.name=username
user.email=username@cognizant.com
```

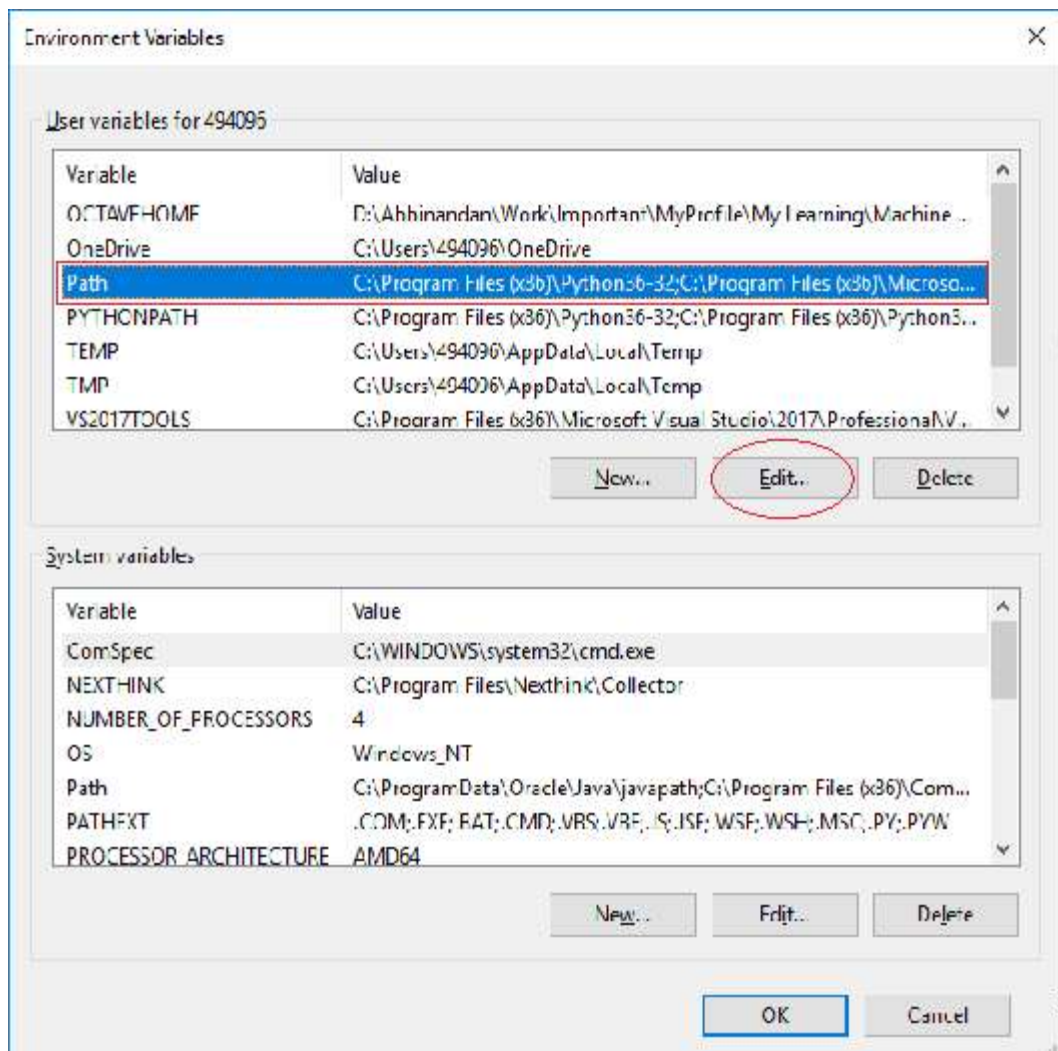
Step 2: Integrate notepad++.exe to Git and make it a default editor

1. To check, if notepad++.exe execute from Git bash

```
$ notepad++
bash: notepad++: command not found
```

If Git bash could not able to recognize notepad++ command that implies notepad++.exe is not added to the environment path variable.

To add path of notepad++.exe to environment variable, go to control panel -> System -> Advanced System settings. Go to Advanced tab -> Environment variables -> Add path of notepad++.exe to the path user variable by clicking on "Edit"



- Exit Git bash shell, open bash shell and execute

```
$ notepad++
```

Now, notepad++ will open from Git bash shell

- To create an alias command for notepad++.exe, execute

```
$ notepad++.exe bash -profile
```

It will open notepad++ from bash shell, and create a user profile by adding the line in notepad++

```
alias npp='notepad++.exe -multiInst -nosession'
```

4. To configure the editor, execute the command

```
$ git config --global core.editor "notepad++.exe -multiInst -nosession"
```

5. To verify if notepad++ is the default editor, execute the command

```
$ git config --global -e  
hint: Waiting for your editor to close the file... _
```

Here '-e' option implies editor

It will show the entire global configuration as shown below,

```
[user]  
  name = username  
  email = username@cognizant.com  
[core]  
  editor = notepad++.exe -multiInst -nosession
```

Step 3: Add a file to source code repository

1. Open Git bash shell and create a new project "**GitDemo**" by executing the command

```
$ git init GitDemo  
Initialized empty Git repository in D:/Development_Avecto/GitDemo/.git/
```

2. Git bash initializes the "**GitDemo**" repository. To verify, execute the command

```
$ ls -al  
total 8  
drwxr-xr-x 1 1049089 0 Jan 13 11:54 ./  
drwxr-xr-x 1 1049089 0 Jan 13 11:54 ../  
drwxr-xr-x 1 1049089 0 Jan 13 11:54 .git/
```

It will display all the hidden files in the Git "working directory".

3. To create a file "**welcome.txt**" and add content to the file, execute the command

```
$ echo "Welcome to the version control" >> welcome.txt
```

4. To verify if the file “welcome.txt” is created, execute

```
$ ls -al
total 9
drwxr-xr-x 1 494096 1049089  0 Jan 13 12:02 ./
drwxr-xr-x 1 494096 1049089  0 Jan 13 11:54 ../
drwxr-xr-x 1 494096 1049089  0 Jan 13 12:01 .git/
-rw-r--r-- 1 494096 1049089 31 Jan 13 12:02 welcome.txt
```

5. To verify the content, execute the command

```
$ cat welcome.txt
Welcome to the version control
```

6. Check the status by executing

```
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)

        welcome.txt
```

Now the file “**welcome.txt**” is available in Git “working directory”

7. To make the file to be tracked by Git repository, execute the command

```
$ git add welcome.txt
warning: LF will be replaced by CRLF in welcome.txt.
The file will have its original line endings in your working directory
```

8. To add multi line comments, we are opening default editor to comment. Execute the command

```
$ git commit
```

Notepad++ editor will open and to add multi-line comment with default editor

9. To check if local and “Working Directory” git repository are same, execute git status

```
$ git status
On branch master
nothing to commit, working tree clean
```

welcome.txt is added to the local repository.

10. Signup with GitLab and create a remote repository **"GitDemo"**
11. To pull the remote repository, execute

```
git pull origin master
```

12. To push the local to remote repository, execute

```
git push origin master
```

SOLUTION

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR powershell + v [ ] [ ] ... | [ ] [ ] [ ]
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT> cd WEEK-6
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6> mkdir GitDemo

Directory: C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6

Mode                LastWriteTime         Length Name
----                -
d-----          29-07-2026  10:38 AM             GitDemo

● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6> cd GitDemo
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> git init
Initialized empty Git repository in C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo\.git\

```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> echo Welcome To Git > welcome.txt
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> dir

Directory:
C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo

Mode                LastWriteTime         Length Name
----                -
-a----           29-07-2026  10:42 AM             38 welcome.txt
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> Get-Content welcome.txt
```

Welcome
To
Git


```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    welcome.txt

nothing added to commit but untracked files present (use "git add" to track)
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> git add welcome.txt
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   welcome.txt
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> git commit -m "Added welcome.txt"
[master (root-commit) 39b0e15] Added welcome.txt
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 welcome.txt
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> git push
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository using

    git remote add <name> <url>

and then push using the remote name

    git push <name>
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> git push
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\GitDemo> cd ..
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6> cd ..
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT> git add .
warning: adding embedded git repository: WEEK-6/GitDemo
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint:   git submodule add <url> WEEK-6/GitDemo
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint:   git rm --cached WEEK-6/GitDemo
hint:
hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT> git commit -m "WEEK-6 welcome.txt added"
[main 1168c96] WEEK-6 welcome.txt added
 1 file changed, 1 insertion(+)
```

```
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT> git commit -m "WEEK-6 welcome.txt added"
[main 1168c96] WEEK-6 welcome.txt added
 1 file changed, 1 insertion(+)
 create mode 160000 WEEK-6/GitDemo
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 325 bytes | 162.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Sravva192000/DN-JAVA-FSE-REACT.git
   fbfb8aff..1168c96  main -> main
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT> 
```